

Advanced (Habanero)

- Q1.** What is the graph of the inequality $y > 2x - 3$?
(A) A line with a slope of 2 and y-intercept of -3
(B) A line with a slope of 2 and y-intercept of 3
(C) A solid line with a slope of 2 and y-intercept of -3
(D) A dashed line with a slope of 2 and y-intercept of -3
(E) A region above a dashed line with a slope of 2 and y-intercept of -3
- Q2.** Which inequality represents the region below the line $y = 3x + 2$?
(A) $y < 3x + 2$
(B) $y > 3x + 2$
(C) $y \leq 3x + 2$
(D) $y \geq 3x + 2$
(E) $y = 3x - 2$
- Q3.** What is the boundary line for the inequality $x + 2y > 5$?
(A) $x + 2y = 5$, solid
(B) $x + 2y = 5$, dashed
(C) $x - 2y = 5$, solid
(D) $x - 2y = 5$, dashed
(E) $2x + y = 5$, solid
- Q4.** Which of the following is a true statement about the graph of $y \geq -2x + 1$?
(A) The boundary line is dashed
(B) The boundary line is solid
(C) The region is below the boundary line
(D) The region is above the boundary line
- (E) The region includes the x-axis
- Q5.** Which inequality has a solid boundary line?
(A) $y < 2x - 3$
(B) $y \leq 2x - 3$
(C) $y > 2x + 1$
(D) $y \geq 2x + 1$
(E) $x + y > 4$
- Q6.** The inequality $y > x - 2$ represents which region?
(A) Below a dashed line
(B) Above a dashed line
(C) Below a solid line
(D) On and below a solid line
(E) On and above a solid line
- Q7.** What type of line is the boundary for $x + y \geq 3$?
(A) Dashed
(B) Solid
(C) Horizontal
(D) Vertical
(E) Diagonal
- Q8.** Which inequality has the same graph as $y > 2x - 4$?
(A) $y \geq 2x - 4$
(B) $y < 2x - 4$
(C) $y \leq 2x - 4$
(D) $y = 2x - 4$
(E) $x > 2y + 4$

Open Ended Questions (FRQ)

Q9. Graph the inequality $y < -x + 2$ and identify the boundary line.

Q10. Write an inequality that represents the region below the line $y = 3x - 2$ and graph the inequality.

Chef's Tips

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- Tip 1: Emphasize the difference between solid and dashed boundary lines.
- Tip 2: Stress the importance of identifying the correct region for each inequality.
- Tip 3: Remind students to always check their work.